Connected Vehicle Telematics Platform

AWS Infrastructure Diagram

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Edge Device

Edge Device:
Vehicle CAN bus interface that decodes OBD-II messages and publishes telemetry to IoT Core.
- MQTT via Basic Ingest
- X.509 certificate auth (mTLS)
- Local buffering during connectivity loss

AWS Cloud

Global & Edge Services

Route53

Route53:
- Hosted Zone: fleet.example.com
- Alias record → CloudFront distribution
- Routing: Simple
- Health checks: API GW + CloudFront (calculated aggregate)
- DNS validation records for ACM

ACM

ACM Certificate:
- Wildcard cert for the domain
- Auto-renewal via R53 DNS
- (us-east-1)

CloudFront (CDN)

CloudFront (CDN):
- TLS termination point
- OAC restricts S3 to CloudFront only
- API path routes to API GW origin
- Signed URLs for S3 media (dashcam, reports)

WAF

WAF:
- AWS Managed Rulesets
- Rate limit: 500 req per IP in 5min
- Geo-restricts to NA only
- Attached to CloudFront

Regional AWS Services

API Gateway Layer

API Gateway

API Gateway (HTTP API):
Authorizers:
- Cognito JWT authorizer for API requests
- `access-token-authorizer` for resource-scoped requests

Ingestion Pipeline

IoT Core

IoT Core:
Managed MQTT broker. Forwards vehicle telemetry to Kinesis via rule action.
- Basic Ingest: $aws/rules/{rule-name}
- Per-vehicle X.509 certificate auth
- DLQ alarm on Kinesis delivery failure

Kinesis Data Stream

Kinesis Data Stream:
Stream for all vehicle telemetry data.
- On-demand (auto-scaling shards)
- 24hr retention
- KMS (CMK)
- Transition to provisioned at larger scale

Kinesis Firehose

Kinesis Firehose:
Archival pipeline for vehicle telemetry.
- Receives batched records from Consumer via PutRecordBatch
- Delivers to S3 in Parquet format
- 5KB batching optimization applied

Partitioned:
- `{type}/pseudoVIN/yyyy/MM/dd/HH`

Async Processing & Queues

SQS

SQS Queues:
- trip-processing
  - Visibility: 120s / Retention: 14d
- geofence-check (FIFO)
  - Visibility: 30s / Retention: 1d
- dashcam-processing
  - Visibility: 900s / Retention 4d

DLQ

Dead Letter Queues:
- trip-processing-dlq - Captures jobs after 3 failed attempts
  - 14d retention
- iot-core-dlq - Captures Kinesis failed deliveries
  - 14d retention

Enrichment & Analytics Services

Location Services

Location Services:
- Geofence boundary evaluations
- Dashboard mapping and routing
- ~2M evaluations/mo at 10K vehicles

Athena

Athena:
Serverless SQL engine that `api-handler` uses to query historical analytics.
- Processes breadcrumb, telemetry event and alerts, charging sessions, and security events

Identity, Secrets & Configuration

Cognito

Cognito:
- Pre-token generation trigger and injects custom claims
- Manages 5 RBAC roles:
  - owner, admin, manager, driver, viewer
- Temporary access tokens (15min)

KMS

KMS:
- 2 CMKs: Data (S3/SNS) + Telemetry (Kinesis)
- Auto-rotation annually
- Bucket keys enabled
- Multi-region keys deployed (DR)

SSM

Parameter Store:
Non-sensitive runtime config management
- Stores ECS and Lambda resource environment variables
- Config changes take effect upon subsequent resource restart

Secrets Manager

Secrets Manager:
Stores and manages secret data
- cognito secrets
- oem-api secrets
- vin-hmac-key
- access-token-key
- `influxdb-api-tokens`

Observability & Security

SNS

SNS:
- Topics:
  - critical-alerts: user-facing alerts
  - security-alerts: admin-facing alerts
- KMS encrypted

EventBridge

EventBridge Rules:
- Stale Trip Cleaner = every 5min
- Cognito Userpool Export = Daily at 0200 UTC
- InfluxDB Predictive Maintenance Analytics = every hour
- InfluxDB S3 Buckup = Daily at 0200 UTC

CloudWatch

CloudWatch:
- Zero-tolerance alarm on sensitive operations
- Threshold alarms on unauthorized API requests and sign-in failures
- VPC Flow Logs enabled

CloudTrail

CloudTrail:
- Logs all AWS API activity
- S3 data events for sensitive buckets
- Log file integrity validation
- S3 bucket has object lock + deny delete policies enabled

GuardDuty

GuardDuty:
- Managed threat detection
- Sources: CloudTrail, VPC Flow Logs, DNS
- S3 protection enabled (data event monitoring)
- Archives findings severity > 7
- Alerts findings via SNS

Elastic Container Registry (ECR)

ECR

Elastic Container Registry:
Stores container images
- Repos: `consumer`, `sse-server`
- Immutable tags
- Retains only the last 10 images
- Enhanced Scanning (Inspector)
- Critical findings block deployment

DynamoDB Tables

vehicle-live
Stores the latest vehicle state by continually overwritting data.
- Global Table (us-west-2 replica)
- On-demand
- PITR / Delete Protection
- High-volume

trip-history
Stores completed trip summaries with references to S3 GZIP path.
- Global Table (us-west-2 replica)
- On-demand
- 1 write per trip (low-volume)
- PITR / Delete Protection

vin-mapping
Vin <-> pseudoVin resolution.
- Global Table (us-west-2 replica)
- On-demand
- Restricted to IAM admin only
- Zero-tolerance CloudTrail and Alarm on any access
- PITR / Delete Protection

oem-tokens
OAuth tokens for OEM API access.
- Global Table (us-west-2 replica)
- On-demand
- PITR / Delete Protection

fleet-operations
Operational data including reports, maintenance info, and configs scoped to an org.
- Global Table (us-west-2 replica)
- On-demand
- PITR / Delete Protection

command-audit
Audit trail for every vehicle command executed.
- Global Table (us-west-2 replica)
- On-demand
- 90d TTL
- PITR / Delete Protection

organization
All org data including tenant metadata, members, drivers, vehicles, and alerts.
- Global Table (us-west-2 replica)
- On-demand
- PITR / Delete Protection

S3 Buckets

cloudtrail-logs
Dedicated bucket for CloudTrail audit log archival.
- KMS (CMK)
- S3 365d compliance object lock
- Versioned
- Block Public Access

dashcam-media
Stores raw and compressed dashcam media.
- KMS (Customer-Managed)
- Versioned
- CloudFront OAC Policy

fleet-reports
Stores generated fleet report exports.
- KMS (Customer-Managed)
- Versioned

telemetry-datalake
Firehose destination for telemetry archive.
- KMS (Customer-Managed)
- Versioned
- Glacier IR at 1 year
  - 2 years for security events

trip-artifacts
Stores gzipped trip artifacts.
- KMS (Customer-Managed)
- Versioned
- CloudFront OAC Policy

web-app
Hosts the static frontend served through CloudFront.
- AES-256 (SSE-S3)
- Versioned
- CloudFront OAC Policy

cognito-exports
Daily Cognito user pool export for DR pipeline.
- AES-256 (SSE-S3)
- CRR to us-west-2
- 30-day lifecycle expiration

VPC (2 AZs)

Public subnet

NAT Gateway

NAT Gateway:
- Multi-AZ
- Utilized for private subnet access to AWS services (**except Kinesis**).
- Only internet egress is secure-sign-proxy to OEM API endpoint.

NLB

Network Load Balancer (NLB):
Routes SSE traffic to Fargate SSE tasks in private subnet.
- Listener: port 3000
- Connection idle timeout: 350s

VPC Endpoints

VPC Endpoints:
- DynamoDB Gateway Endpoint
- S3 Gateway Endpoint
- Kinesis Data Stream Interface Endpoint
- Kinesis Firehose Interface Endpoint

Private subnet

ECS Cluster (Fargate Services)

consumer-service
Polls Kinesis Data Stream for new vehicle telemetry, pseudonymizes the VIN and fans it to DDB, Firehose, S3, SQS, TimeStream and SNS.

sse-streaming-service
Subscribes to ElastiCache Valkey and pushes real-time updates to user's dashboards via server sent events (SSE).

Upon SSE failure, the frontend falls back to DynamoDB polling until SSE recovers. Clients auto-reconnect.

Pub/Sub Broker

ElastiCache Valkey

ElastiCache Valkey:
- Pub/sub broker between the Consumer and SSE server.
- Multi-AZ
- cache.t4g.micro instances

TimeStream

TimeStream (InfluxDB)

TimeStream (InfluxDB):
Predictive maintenance analytics engine.
- Receives telemetry from Consumer via Line Protocol
- db.influx.medium
- 10 buckets, 90-365d retention

Lambda Functions (VPC-Attached)

api-handler
API handler behind API Gateway. Handles CRUD for vehicles, trips, organizations, and fleet operations.

oem-command-proxy
Signs and forwards vehicle commands to OEM API endpoints. Writes audit records for every command.

trip-processor
Computes trip summary metrics, safety events, and driving scores from archives. Triggered by DynamoDB Streams.

geofence-evaluator
Evaluates vehicle positions against configured geofence boundaries and publishes alerts.

dashcam-processor
Transcodes raw dashcam clips to web ready format and generates thumbnails. Triggered by SQS.

stale-trip-cleanup
Detects and gracefully completes orphaned trips that lost connectivity mid-drive. Runs on a 5 minute schedule.

predictive-maintenance
Runs queries against InfluxDB to detect component degradation trends. Runs hourly via EventBridge.

token-generator
Issues temporary tokens and CloudFront signed URLs for sensitive assets like dashcam media.

token-authorizer
Validates tokens on protected API routes. Used as an API Gateway custom authorizer.

cognito-pre-token
Injects custom claims (organizationId, role, pseudoVINs) into Cognito JWTs.

cognito-dr-export
Daily DR export of the Cognito user pool to S3.

cognito-dr-import
Import of user pool backup from S3 into us-west-2 for DR.